# Varun Ravi

CS @ NUS · Game & systems developer

## CONTACT


Singapore

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## EDUCATION

**National University of Singapore**
BComp Computer Science
Aug 2026 – May 2030 (expected)

**Tampines Meridian Junior College**
GCE A-Levels
2022 – 2023

## SKILLS

**Languages**
C, C++, Rust, Python, x86-64 Assembly; also Java, JavaScript, GLSL

**Game development**
OpenGL, SDL2, pygame, Godot, Tiled

**Tools**
Git, GDB, Linux, CMake, Make, QEMU

## AWARDS

- **Overall Champion, NUS CFG x Garena Ultimate Hackathon 2026** — Make Jelly, a Roblox incremental tycoon game by a 3-person team. Contributed game concept ideation and playtesting: pitched early concepts and flagged the missing background music, which was then added.
- **Best Soldier of the Month**, Singapore Army, May 2025
- **Honourable Mention**, Singapore Mathematical Olympiad (SMO)

## GAMES

- **Samsara Tiles** (C++, SDL2) — GMTK Game Jam 2025, built in 3 days in a team of 2. A 2D platformer whose level changes on every loop. Wrote the custom engine and a tile-based level editor.
- **Maze Runner** (C++, OpenGL) — A timed 3D maze game built on my own engine. Limited view distance implemented with depth-buffer post-processing.
- **Graph It!** (Python, pygame-ce) — Mini Jam #183, built in 3 days. A puzzle game about matching functions to graphed points.
- **glgame2d** (C++, OpenGL) — A 2D game framework with Tiled tilemap loading, audio and Catch2 tests.

## EXPERIENCE

**Lead Developer** — Dec 2024 – Mar 2026
Singapore Army, 6AMB Innovation Project Team

- Took over as sole developer after the original leads left; later recommended a new teammate and helped onboard them, advising on code structure
- Led development of a new front-end for FIRMS, the first portable AI-powered marksmanship trainer for the SAR21, giving trainers and trainees real-time diagnostic feedback
- Improved the machine learning models that identify flaws in soldiers' firing fundamentals

**Intern** — Dec 2023
Rohde & Schwarz

- Built Python scripts with a tkinter GUI that parse and update HTML test reports automatically, replacing manual edits and speeding up product testing

## PROJECTS

- **x86-64 Kernel** (C, x86-64 Assembly) — A hobby kernel booted with Limine. Features a physical frame allocator, 4-level paging, a kernel heap, APIC interrupts, a preemptive round-robin scheduler and an interactive shell.
- **Brush Converter** (Python) — Converts brushes between Procreate, GIMP, Medibang, Clip Studio Paint and Photoshop. Reverse-engineered undocumented formats. Ships as a prebuilt GUI app with nearly 200 downloads and 10 GitHub stars from digital artists.
- **MathFP** (Rust) — A functional programming language with math-like syntax. Handwritten lexer and parser with line and column error reporting, plus transpilation to LaTeX.
- **Also built:** a CHIP-8 emulator in C and a multithreaded ray tracer in C++.

## OPEN SOURCE

- **C3 compiler** — Added a compile error for unparenthesized assignments in `if` conditions, catching = vs == typos, plus a regression test (c3c #2729, merged).